

Balaji Rao

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Education

Stevens Institute of Technology, Hoboken, NJ – ABD

(Expected) October 2026

Ph.D., Systems Engineering

Dissertation Area: Large Language Models (LLMs), LLM Reasoning, neurosymbolic AI, and Systems Reliability

Title: Enhancing Reliability of LLM Generated Content Through Structure and Machine Checks

Advisor: Dr. Carlo Lipizzi

Stevens Institute of Technology, Hoboken, NJ

May 2022

Master of Engineering, Engineering Management

Graduate Certificates: Data Exploration and Visualization for Risk and Decision-Making; Systems Engineering

BNM Institute of Technology, Bengaluru, India

October 2020

Bachelor of Engineering, Electronics and Communication Engineering

Profile

Ph.D. candidate in Systems Engineering working on reliable LLM systems, neural theorem proving, and verifier-backed evaluation. My research builds structure-and-check pipelines where models generate formal artifacts that are checked by proof assistants, SMT solvers, and executable evaluators. I am interested in using formal methods for feedback environments to study model reasoning, specification fidelity, and code correctness.

Skills

Programming Languages: Python, OCaml, Rust, Lean, SMT-LIB

Frameworks and Libraries: PyTorch, TensorFlow, scikit-learn, Hugging Face, NumPy, Pandas

Machine Learning: LLM fine-tuning, Reinforcement Learning, Neural Theorem Proving, Autoformalization

Tools and Platforms: Isabelle/HOL, HOL Light, AWS Bedrock, Amazon Quick, Claude Code, Kiro, Cursor

Experience

Applied Science Intern - Amazon Quick (AWS) [In Progress]

May 2026 – August 2026

- Integrating formal methods into Amazon Quick to improve the reliability of generated answers through structured specifications, verification-backed checks, and trustworthy agent-generated content.

Applied Science Intern - Automated Reasoning (Amazon Web Services)

May 2025 – August 2025

- Built an LLM-assisted HOL Light/s2n-bignum tool for tactic explanation and suggestion, integrating Bedrock, SageMaker, Kendra, S3, and an MCP server for IDE-based prover interaction.
- Prototyped a verifier-in-the-loop theorem-proving environment where LLMs propose HOL Light tactics, receive proof-checking feedback, and retry failed proof attempts; added evaluation artifacts and migrated experiments to a locally hosted model to reduce token costs and support rapid iteration.
- Enhanced verification workflows by automating formal correctness proofs and generating natural-language descriptions of custom tactics, reducing manual effort for software verification engineers.

Data Science/Data Engineering Intern - Johnson & Johnson

May 2024 – August 2024

- Developed machine learning models to analyze and reduce content fatigue, enhancing healthcare professionals' (HCPs) engagement with promotional emails.
- Implemented a Hidden Markov Model (HMM) for probabilistic predictions of email engagement, utilizing a feature matrix that included temporal data. Integrated use of Gen AI solutions to leverage large language models (LLMs) like Llama-2 to optimize content, improving messaging outcomes.
- Introduced new predictive analytics metrics to provide deeper insights into content fatigue, complementing traditional email engagement metrics.

Research Assistant – Stevens Institute of Technology, Hoboken, NJ

July 2021 – Present

- Co-developed *s2n-bignum-bench*, a HOL Light benchmark of 2,301+ (evolving) proof obligations distilled from the AWS *s2n-bignum* verified cryptography library to evaluate ISA-aware, bit-precise LLM reasoning on low-level code. Active contributor to AWS's *s2n-bignum* Project.
- Designed *ProofSeek*, an LLM-based neural theorem proving system that formalizes natural-language policies into Isabelle/HOL proofs. Fine-tuned models using a two-stage training regime and evaluated performance using the miniF2F benchmark and real-world cloud security policy datasets (e.g., AWS S3). Deployed a proof assembly module (ProofAug) and applied symbolic verification techniques to extend prior work on formal code validation.

- Developed structure-and-check LLM pipelines that convert free-form specifications into machine-checkable artifacts, using typed logical representations, verification plans, SMT-based checks, and executable SysML models to improve reliability and catch contradictions early.
- Designed and implemented an automated theorem-proving framework using Isabelle/HOL for verifying policy logic in safety-critical domains. Combined symbolic reasoning with reinforcement learning to improve formal proof generation.
- Proposed and began developing a framework for self-refining LLM systems using reinforcement learning reward tuning to optimize output coherence and factual accuracy across applied generative tasks.
- Constructed a multimodal classification framework combining EEG and facial expression data to infer user preferences. Evaluated performance across several supervised learning algorithms.

Selected Projects

Vibes to Verification: Evidence-Gated LLM Workflows

May 2026

Developed a verification-in-the-loop framework for agentic LLM systems that uses pre-hoc verification plans, post-hoc diagnostics, and formal evidence gates to separate plausible code generations from checkable guarantees. Applied the framework to C-to-Rust migration workflows with Kani-backed equivalence checks, safety gates, and explicit trust-boundary analysis.

s2n-bignum-bench: Low-Level Theorem Proving Benchmark for LLMs (HOL Light)

December 2025

Packaged 2,301 machine-checkable proof problems derived from the AWS *s2n-bignum* cryptography library into isolated context–query tasks for neural theorem proving evaluation, targeting low-level, bit-precise reasoning. Built an end-to-end offline evaluation pipeline that compiles and checks submitted tactics against shipped object-code artifacts, and added integrity defenses and a lightweight category taxonomy.

Neural Theorem Proving with LLMs: The ProofSeek Framework

March 2025

Developed *ProofSeek*, a neural theorem proving framework combining symbolic reasoning with large language models. The system translates natural-language policies into formal logic, generates Isabelle/HOL proofs using a fine-tuned LLM, and constructs verifiable proofs through a heuristic module (ProofAug). Trained using supervised fine-tuning and reinforcement learning with verifier-derived rewards (GRPO); evaluated on miniF2F and applied to AWS S3 access control policy verification.

Autoformalization for MBSE: LLM $\rightarrow$ SMT $\rightarrow$ SysML

2025

Built a structure-and-check pipeline that converts natural-language requirements into typed logical representations, SMT-LIB constraints checked with Z3, and executable SysML artifacts. The pipeline uses SAT/UNSAT evidence and model compilation to catch ambiguity, contradictions, and malformed requirements before human review.

Formal Verification of Policy Code using Isabelle/HOL

April–May 2024

Implemented a verification pipeline that translates LLM-generated policy code into formal logic using the Isabelle/HOL proof assistant and Isar syntax. The project focused on formalizing cloud-based access policies (e.g., AWS S3) and exploring the intersection of symbolic reasoning with language model output.

SSE-EduBot: Department-Wide Tutoring System using LLMs

2023–2025

Designed and deployed an educational chatbot prototype to assist graduate students at Stevens Institute of Technology. Built using Llama 2, LangChain, and Pinecone, the system is aligned with the Systems Engineering curriculum and provides course-aware, domain-specific responses by leveraging instructional content and syllabi.

Publications, Conference Papers & Presentations

- s2n-bignum-bench: Low-Level Theorem Proving Benchmark for LLMs, AIPV 2026
- *Vibes to Verification*, High Confidence Software and Systems Conference, HCSS 2026 Presentation (Poster)
- *Steve: LLM Powered ChatBot for Career Progression*, AAI 2026 System Demonstration Track
- *SSE-EduBot: Department-Wide Tutoring System Using a Large Language Model*, FIE 2025
- *Neural Theorem Proving: Generating and Structuring Proofs for Formal Verification*, NeSy 2025
- *Anatomy of an AI Economy*, IEEE ISSE 2024
- *Measuring Resilience in Multimodal Transportation Systems*, TRB 2024 (Poster)
- *Identification of Variables Impacting Cascading Failures in Aerospace Systems*, CSER 2024
- *A Game-Theoretic Approach for Validator Selection in PoS Blockchains*, ICoABCD 2023

Fellowships and awards

- *Amazon Post-internship Fellowship* – Financial Support in addition to \$10,000 in AWS credits.
- *Future AI Leaders Program* – The AI Summit New York.
- *Summer School on Formal Techniques (2026)* – SRI (formerly known as Stanford Research Institute).